WEB PROGRAMMING

MIDSEMESTER EXAM

SEMESTER 2

BIS2/BIT2

1. Question 1 – Web Technologies. (25 marks)
   a. The terms “internet” and “world wide web” are often used interchangeably, but they refer to different technologies. Define each term and discuss the key differences between them, highlight their roles and functions in the realm of web technologies. (2 marks)
   b. Explain the roles of any two application layer protocols and one transport layer protocol that are crucial for we applications, highlighting their functions and significance. (6 marks)
   c. In mobile-first design, the process starts by optimising the layout for smaller screens and then progressively enhancing it for larger screens. Explain how media queries play a key role its implementation. (4 marks)
   d. In each response to an HTTP request, a server includes a status code which indicates the result of the request. Explain the meaning of the following response codes.
      i. 201. (1 mark)
      ii. 401. (1 mark)
   e. Explain what tools you would implement in the following guidelines aimed at making web applications to be engaging and attractive.
      i. Incorporate interactive elements. (2 marks)
      ii. Use user generated content. (2 marks)
      iii. Use multimedia elements. (2 marks)
   f. The invention of the world wide web (www) is attributed to Tim Berners-Lee, who developed a set of foundational technologies that form the core of the web. Briefly discuss the score technologies. (5 marks)

2. Question 2 – HTML and CSS. (25 marks)
   a. HTML elements are either inline or block elements, however they could be transformed to inline-block elements. Discuss inline-block elements, your discussion must include their characteristics and scenarios which you might need to use them. (4 marks)
   b. vh together with vw and percentage (%) are some of the measurements units you can apply to HTML elements to specify their size. What are the disadvantages of using % over vh & vw when specifying the size of elements. (4 marks)
   c. perfect centering in CSS refers to aligning an element both horizontally and vertically within its parent container, regardless of the container’s size. Write CSS

styles that will make a div component with the class child to be perfectly centered given that its parent is the body. (6 marks)

```
.child {
    width: 200px;
    height: 160px;
    background-color: #1498cb;
}
```

d. Discuss the difference between transitions and animations in CSS, your discussion must include where it is necessary to use one over the other. (4 marks)
e. Discuss how you would use flexbox CSS to layout components like in the following figure. (6marks)

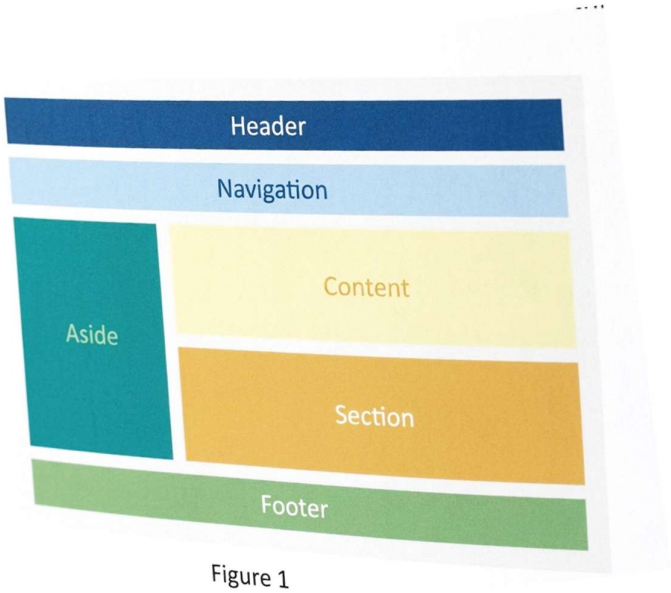


Figure 1

3. Question 3 – JavaScript (25 marks)
    a. What is the Document Object Model and how important is it web application development? (4 marks)
    b. JavaScript is a standardised programming; the standardisation is done by the European Computer Manufacturers Association (ECMA) in specifications called ECMA scripts (ES). Discuss any three features that the body introduced in ES6. (6 marks)
    c. Discuss how you would use event handling to validate user input data in a form before submission. (2 marks)
    d. Describe how you would implement a script which hides and reveals the content of a paragraph when a user click the heading of the paragraph. (4 marks)
    e. Asynchronous programming is a technique that allows programs to execute tasks concurrently. Discuss the different way of implementing asynchronous programming in JavaScript. (3 marks)

f. The following code is showing a definition and a call to it. The function is basically making a GET request to an API endpoint to get some data using fetch API. Use this information to answer the following questions.

```
<script>
  async function fetchData(url) {
    const response = await fetch(url);
    const data = await response.json();
    console.log(data);
  }

  fetchData("https://jsonplaceholder.typicode.com/todos");
</script>
```

i. When the data has been fetched, the code is simply logging to the console. Assuming that the data is a paragraph, write code that will display the data in a non-existent p element in the document. (4 marks)

ii. What are the important HTTP request headers that are key when interacting with an API endpoint when sending data using POST method. (2 marks)

4. Question 4 – Node.js. (25 marks)
   a. Express.js is a lightweight and flexible web application framework for Node.js. it provides a robust set features to build web applications and APIs quickly and efficiently. Explain any three features of the express framework. (3 marks)
   b. Describe the steps that you would use to implement a route to "/home" through a get request and return html content "<h1>hello world</h1>". (4 marks)
   c. Mention any two package managers that can be used to install packages in a Node.js application. (2 marks)
   d. Describe the steps that are involved in the implementation and use of local modules in Node.js. (6 marks)
   e. Data validation and sanitisation are essential processes in web development to ensure that incoming data both meets certain criteria and free from malicious content. Explain the differences between the two approaches to data ensuring data quality. (4 marks)
   f. Session and cookies are two approaches that can be used to track users in a web application. Discuss the differences between the two approaches focusing on the strength and weaknesses. (4 marks)
   g. In express.js, middleware refers to functions that execute during the lifecycle of a request to the server. Describe how you would use the app.use((req,res,next)=>{}) middleware, to implement a logging mechanism. (2 marks)